

Critical Outbreak Size for Disease Importation into the United States: A Threshold Analysis Framework

Research Pipeline and Student Brief

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Overview and Motivation

Context. The FIFA World Cup 2026 importation risk analysis (Herrera-Diestra, in preparation) estimates the probability of at least one imported infectious disease case reaching each US host city, given observed outbreak sizes in source countries. That model answers the question: “*Given today’s incidence, what is the risk?*”

This project asks the inverse question:

*For a given source country and disease, how large does an outbreak need to be before at least one imported case is expected to arrive in the United States?
And, given the current outbreak trajectory, **when** does that threshold get crossed?*

This reframing converts the importation model into an **early-warning tool**: instead of retrospectively estimating risk from observed data, it provides a *critical threshold* C_c^* that public health authorities can monitor in real time. When surveillance reports that a country’s case count has crossed C_c^* , the probability of at least one US importation has become non-negligible.

Why this matters. A single imported case of a high-severity pathogen (e.g., Ebola, Marburg, novel influenza) can trigger enormous public health responses. Knowing in advance *how large* an outbreak in country c must be to generate even a 5% importation probability allows:

- Proportionate pre-positioning of hospital resources at high-volume entry airports.
- Evidence-based decisions on enhanced traveller screening.
- Communication to public health officials about when to escalate monitoring vs. when risk remains negligible.

Relationship to the WC paper. The mathematical framework here is identical to the WC importation model. All base data (travel volumes, population, disease incidence) and all code are inherited from that project. This analysis extends it in a new direction and is intended as a standalone follow-up paper.

1 Mathematical Framework

1.1 The Poisson Importation Model (Inherited)

The WC analysis models disease importation as a Poisson process. If $\Lambda_{c,d}$ is the expected number of imported cases of disease d from country c arriving in the US per month, the probability of observing at least one importation is:

$$P(X_{c,d} \geq 1) = 1 - e^{-\Lambda_{c,d}} \tag{1}$$

The importation intensity is:

$$\Lambda_{c,d} = N_c \cdot I_{c,d} \cdot p_d = N_c \cdot \frac{\rho_d \cdot C_{c,d}}{P_c} \cdot p_d \quad (2)$$

where:

- N_c — monthly travelers from country c to the US (COR/I-94 data, June average).
- $C_{c,d}$ — reported cases of disease d in country c per month.
- P_c — population of country c .
- ρ_d — reporting adjustment factor for disease d (corrects for under-reporting).
- p_d — probability that an infected individual travels internationally while infectious.

1.2 Inverting the Model: Critical Outbreak Size

Setting Equation 1 equal to a threshold $\theta \in (0, 1)$ and solving for $C_{c,d}$:

$$1 - e^{-\Lambda_{c,d}} = \theta \implies \Lambda_{c,d} = -\ln(1 - \theta) \quad (3)$$

Substituting Equation 2 and solving for the **critical case count** $C_{c,d}^*$:

$$C_{c,d}^* = \frac{-\ln(1 - \theta) \cdot P_c}{N_c \cdot \rho_d \cdot p_d} \quad (4)$$

Interpretation. $C_{c,d}^*$ is the number of active monthly cases in country c at which the probability of exporting at least one case of disease d to the US reaches θ . It should be computed for multiple threshold levels, e.g.:

Threshold θ	$-\ln(1 - \theta)$	Interpretation
0.05	0.051	“Emerging concern”
0.10	0.105	“Low risk”
0.50	0.693	“Even odds”
0.95	2.996	“Near-certain importation”

Note that $C_{c,d}^*$ varies dramatically across countries and diseases. A country with large travel volume to the US (e.g., Brazil, Mexico) has a much lower threshold than a country with minimal travel (e.g., DRC).

1.3 Temporal Component: Time to Threshold

If an outbreak is growing, the number of cases at time t (in days) follows an exponential growth model in the early phase:

$$C(t) = C_0 \cdot e^{r \cdot t} \quad (5)$$

where C_0 is the observed case count at time $t = 0$ (today) and r is the epidemic growth rate. The growth rate is related to the basic reproduction number R_0 and the mean serial interval T_s :

$$r \approx \frac{\ln R_0}{T_s} \quad (6)$$

Setting Equation 5 equal to $C_{c,d}^*$ and solving for t :

$$t^* = \frac{1}{r} \ln \left(\frac{C_{c,d}^*}{C_0} \right) \quad (7)$$

Interpretation. t^* is the number of days from today until the outbreak is expected to cross the importation risk threshold, given the current trajectory. If $C_0 > C_{c,d}^*$, the threshold has already been crossed ($t^* < 0$).

Important caveat. The exponential model is a simplification valid only for the early epidemic phase (before control measures, depletion of susceptibles, etc.). For more realism, this can be replaced by a SEIR model, but the exponential approximation is sufficient for a first-pass early-warning tool.

1.4 Aggregate US Importation Risk

The framework above can be applied to individual US entry airports by replacing N_c with $N_{c,v}$ (arrivals from country c to city v) using T-100 routing fractions. This allows city-level threshold curves analogous to the WC analysis.

2 Diseases to Include

The analysis should cover at minimum the four diseases from the WC paper, plus Ebola as a high-severity case study that motivates the project. Parameters are listed below; see Table 1 for sources.

Note on R_0 uncertainty. The R_0 estimates above are literature ranges. The temporal analysis should include a sensitivity analysis varying R_0 within its plausible range.

3 Data Requirements

Most data are already available from the WC project. New data needed are marked with $\star$.

Table 1: Disease parameters for the threshold analysis. ρ_d = reporting adjustment; p_d = travel-while-infectious probability; R_0 and T_s are used in the temporal model. Ebola parameters are for the Bundibugyo strain (May 2026 DRC outbreak).

Disease	ρ_d	p_d	R_0	T_s (days)	Incidence source
Dengue	0.3	0.50	2.0–6.0	20	WHO monthly
Malaria	0.2	0.30	1.1–3.0	30	WHO annual
Measles	0.6	0.05	12–18	12	WHO annual
Pertussis	0.3	0.70	5.5–17	14	WHO annual
Ebola	0.8	0.01	1.5–2.5	15	WHO situation reports
<i>Optional extensions (Phase 2):</i>					
Mpox	0.5	0.30	1.5–3.0	9	WHO/CDC
Cholera	0.3	0.20	1.5–4.0	5	WHO

3.1 Already Available (WC Project)

- **Country populations:** Data/population2020.csv (2020 baseline; acceptable for threshold calculations).
- **Monthly COR arrivals by country:** Data/Monthly_Arrivals_Country_of_Residence_COR_1.csv — use mean June arrivals as N_c .
- **T-100 routing fractions:** Data/t100_routing_fractions.csv — for city-level extension.
- **Dengue incidence (monthly, country-level):** Data/dengue-global-data-2025-12-10.xlsx
- **Malaria incidence (annual, per 1,000):** Data/Malaria_National_Unit_data.csv
- **Measles incidence (annual, per 1,000,000):** Data/Measles reported cases and incidence *.xlsx
- **Pertussis incidence (annual, per 1,000,000):** Data/Pertussis reported cases and incidence *.xlsx
- **Country–region mapping:** derived from COR data in importationRisk_main.R.

3.2 New Data to Obtain (★)

- ★ **Ebola incidence data.** Download WHO Disease Outbreak News (DON) and situation reports for all Ebola outbreaks since 2000.
URL: <https://www.who.int/emergencies/disease-outbreak-news>
Format needed: country, date, confirmed cases, deaths. For ongoing outbreaks, use the CDC situation summary:
<https://www.cdc.gov/ebola/situation-summary/index.html>
- ★ **R_0 and serial interval estimates (literature).** Compile a table of R_0 and T_s estimates from the published literature for each disease. Suggested sources:

- Biggerstaff et al. (2014) for influenza.
 - Althaus (2014) for Ebola (R_0 West Africa).
 - WHO Technical Reports for other diseases.
- ★ **Air passenger volumes by country (most recent COR).** The existing COR file covers through mid-2025. If a more recent extract is available from CBP (<https://www.cbp.gov/newsroom/stats/cbp-travel-stats>), download and add to **Data/**.
- ★ **Optional: historical importation events.** Compile a validation dataset of documented disease importations into the US from 2010–2025 (source country, disease, date). Use ProMED (<https://promedmail.org>) and CDC importation reports. This will be used in Section **T4**.

4 Task List

Tasks are numbered and grouped by phase. Each task has a suggested time estimate for a graduate student working part-time.

4.1 Phase 1 — Analytical Framework (Weeks 1–2)

T1. Set up the project structure. Fork or clone the WC repository (https://github.com/jdiestra1977/FIFA_worldCup_2026_risk). Create a new folder **OutbreakThreshold/** with a dedicated R script **threshold_analysis.R**. Load all shared data files from **Data/** — do not duplicate them. *[1–2 hours]*

T2. Implement the critical threshold formula. Code Equation 4 as a function in R:

```
compute_threshold <- function(N_c, P_c, rho_d, p_d,
                             theta = c(0.05, 0.10, 0.50, 0.95)) {
  -log(1 - theta) * P_c / (N_c * rho_d * p_d)
}
```

Apply this to all combinations of countries (from COR data) and diseases (from Table 1). Output: a tidy data frame with columns **Country**, **disease**, **theta**, **C_star**. *[3–4 hours]*

T3. Compute thresholds for all country $\times$ disease pairs. Run **T2** for all countries in the COR dataset and all five diseases. Use the mean annual COR arrivals (not June-only) for N_c here, since this analysis is not WC-specific. Identify the top 20 countries per disease with the *lowest* C^* (i.e., most at-risk source countries). *[2–3 hours]*

T4. Sanity check and validation against known events. For three well-documented US importation events (e.g., 2014 Ebola in Dallas from Liberia; 2019 measles from Israel; 2022 Ebola from DRC/Uganda), check whether the observed outbreak size at time of travel exceeded C^* for $\theta = 0.05$. If the model is well-calibrated, the answer should be “yes” for most cases. *[4–6 hours; requires historical data from T★4]*

4.2 Phase 2 — Temporal Model (Weeks 3–4)

- T5. Implement the epidemic growth model.** Code Equations 5–7. For each disease, use the central R_0 estimate from Table 1 to compute r . Implement a sensitivity wrapper that reruns the calculation for the low and high R_0 bounds. [3–4 hours]
- T6. Apply to the current DRC/Ebola outbreak (case study).** Use the May 2026 CDC situation summary as C_0 : DRC has 347 total suspected/confirmed cases at the time of writing. Compute t^* for $\theta \in \{0.05, 0.50\}$. Compare to dengue in Brazil as a contrast. [2 hours]
- T7. Generate threshold curves over time.** For the top 10 highest-risk country $\times$ disease pairs (identified in T3), plot $P(\geq 1)$ as a function of outbreak size C (x-axis) for a range of 0 to $2 \times C^*$. This produces “risk ramp” curves that visualise how quickly importation probability grows with outbreak size. [3 hours]

4.3 Phase 3 — Figures and Outputs (Weeks 5–6)

- T8. Figure 1: Global threshold map.** Choropleth world map where the fill colour for each country is $\log_{10}(C^*/\text{current incidence})$ for dengue at $\theta = 0.50$. Ratio < 1 (country already above threshold) shown in red; ratio 1–10 in orange; > 10 in green. Use `ggplot2 + sf + country polygons` from `rnaturalearth`. [4–6 hours]
- T9. Figure 2: Threshold ranking table (top 20 per disease).** Faceted bar chart: x-axis = country, y-axis = C^* (log scale), faceted by disease. Annotate current reported incidence as a point on each bar to show how close each country is to the threshold. [3–4 hours]
- T10. Figure 3: Time-to-threshold curves.** For the DRC/Ebola and Brazil/Dengue case studies, plot outbreak size $C(t)$ vs. time (x-axis), with a horizontal dashed line at C^* for each θ level. Shade the region where $C(t) > C^*$ in red. [2–3 hours]
- T11. Figure 4: Sensitivity analysis.** For a representative country $\times$ disease pair, show how C^* changes when ρ_d , p_d , and N_c are each varied $\pm 50\%$ independently (tornado plot or similar). [2–3 hours]
- T12. Figure 5: City-level extension (optional).** Using T-100 routing fractions, compute city-specific thresholds $C_{c,v}^*$ for the top 5 US entry cities and the top 5 source countries per disease. Display as a heatmap analogous to the WC paper. [4–5 hours]

4.4 Phase 4 — Write-up (Weeks 7–8)

- T13. Draft manuscript.** Target journal: *PLOS Neglected Tropical Diseases* or *Emerging Infectious Diseases*. Suggested structure: Introduction, Methods (model + data), Results (Figures 1–4), Discussion, Supplementary tables (full C^* table for all countries $\times$ diseases). Target length: 3,500–4,500 words. [15–20 hours]
- T14. Produce supplementary C^* table.** A CSV/Excel file with columns: Country, disease, theta_05, theta_10, theta_50, theta_95, current_incidence, ratio_to_threshol

This is a practical deliverable that can be shared with public health departments independently of the paper. *[2 hours]*

5 Suggested R Script Structure

The student should create a single self-contained R script `OutbreakThreshold/threshold_analysis.R` with the following sections. This mirrors the structure of `importationRisk_main.R` in the parent project.

```
# 0. Packages
# 1. Working directory and shared data loading
#   (population, COR arrivals, T-100 routing, disease incidence)
# 2. Disease parameters table
#   (rho, p, R0, T_s for each disease)
# 3. compute_threshold() function [Eq. 3]
# 4. compute_time_to_threshold() function [Eq. 5]
# 5. Critical thresholds: all country x disease combinations
# 6. Temporal model: DRC/Ebola and Brazil/Dengue case studies
# 7. Figure 1: Global threshold map
# 8. Figure 2: Threshold ranking (top 20 per disease)
# 9. Figure 3: Time-to-threshold curves
# 10. Figure 4: Sensitivity analysis
# 11. Figure 5: City-level extension (optional)
# 12. Export supplementary table
```

6 Connections to the WC Paper

The student should be aware of and refer to the parent analysis throughout. Key shared components:

- The Poisson framework (Eq. 1–2) is identical.
- The parameter values in Table 1 must match Table 1 of the WC paper to ensure consistency.
- The T-100 routing fractions and COR data are used as-is.
- The country name harmonisation table in `t100_routing_prep.R` should be reused to avoid join failures.
- The WC paper should be cited as the methodological origin of the importation model.

The threshold analysis is **not** limited to WC-qualified countries or to June travel. It uses full annual travel volumes and covers all 180+ countries in the COR dataset, making it a more general tool.

7 Expected Timeline Summary

Week(s)	Milestone	Deliverable
1–2	Analytical framework coded and validated	R script §0–4
3–4	Temporal model + case studies	R script §5–6
5–6	All figures generated	Figures 1–5
7	Supplementary table complete	.csv file
8	First draft manuscript	.tex draft

Total estimated effort: 60–80 hours for a graduate student with intermediate R skills and basic epidemiology background.

8 Questions to Discuss with the PI

Before starting, the student should clarify the following with Prof. Herrera-Diestra:

1. Which diseases to prioritise in Phase 1 (all five, or start with dengue and Ebola as contrasting examples)?
2. Whether to use mean annual COR arrivals or June-specific arrivals for N_c (the former is more general; the latter is more conservative for WC-period applications).
3. Whether the optional city-level extension (Figure 5, Task T12) is in scope for the first submission.
4. Target journal and co-authorship expectations.
5. Access to the parent GitHub repository.